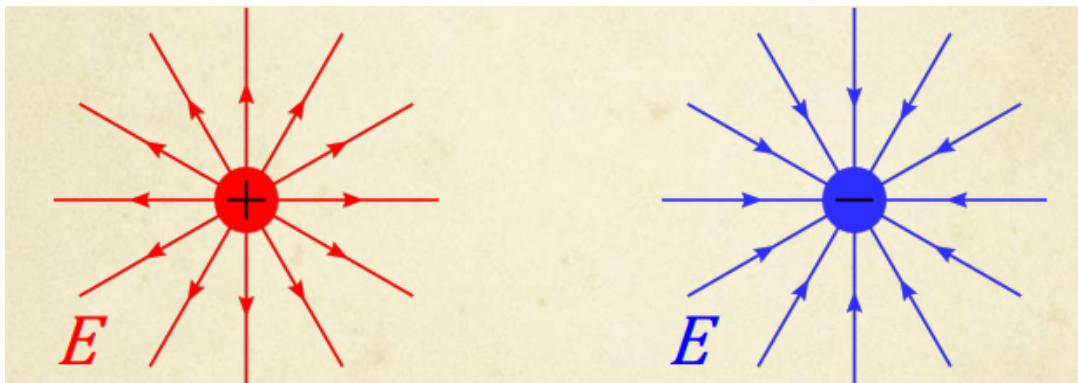


Charged Particles, Conductors and Electric and Magnetic Fields

Stationary Charges

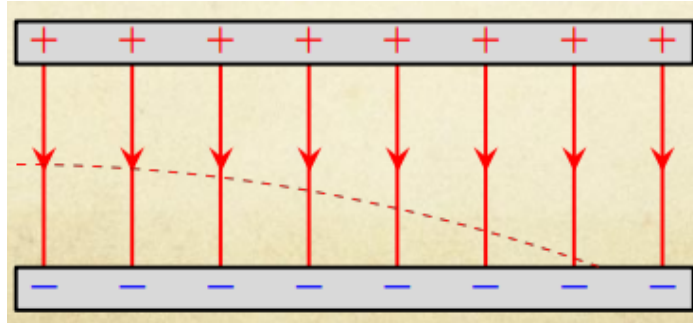
- A stationary charge (q) is surrounded by an electric field (E).
 - The units for q is coulombs (C) and the units for E are volts per metre (Vm^{-1}) or newtons per coulomb (NC^{-1}).



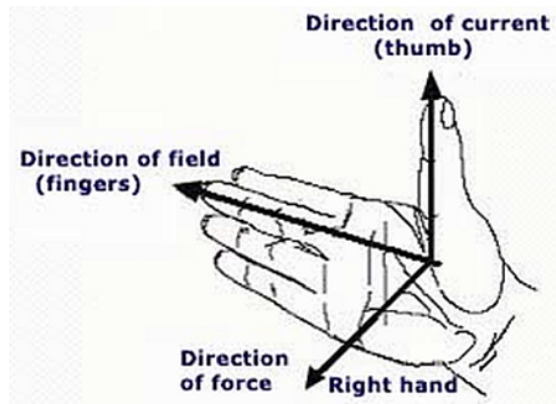
- A stationary charge will experience a force in an electric field, but will not experience a force in a magnetic field.

Charges in an E and B Field

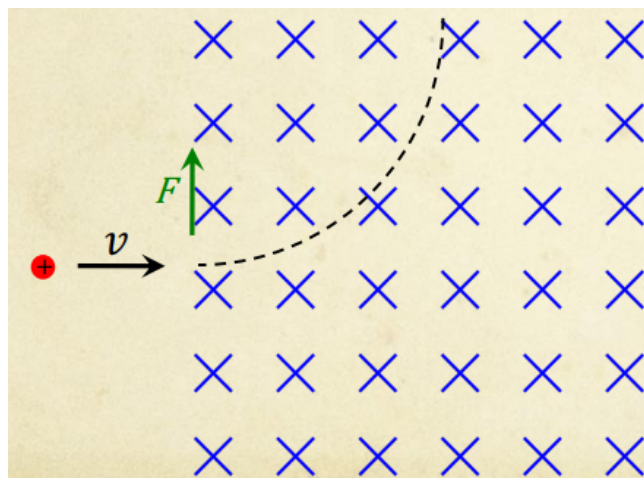
- When a charged particle enters the field between two electric plates, the particle follows a parabolic trajectory.
 - This is because the uniform field applies a constant force, and the particle behaves like a projectile.
- The force applied is in the direction of the opposite sign to the charge.
 - For example, a proton will have a force applied towards the negative plate.



- A moving charge will experience a force in a magnetic field.
 - The force follows the right hand palm rule, where the thumb is in the direction of the velocity of the positive charge, the fingers are in the direction of the magnetic field lines, and the palm is in the direction of the force.



- Because the force is always perpendicular to the velocity, which is always changing direction, the charge will follow a circular trajectory.



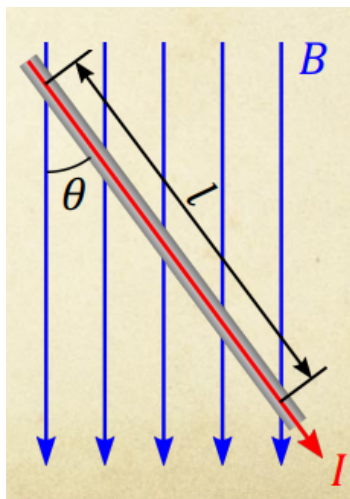
Summary Table of Charges in Different Fields

	Electric Field	Magnetic Field
Stationary Charge	Experiences a force	Does not experience a force
Moving Charge	Experiences a force	Experiences a force
Orientation of Force	Along field lines	Perpendicular to velocity
Equation for Force	$F = qE$	$F = qvB\sin\theta$
Shape of Trajectory	Parabolic	Circular

The Motor Effect

Currents in a Magnetic Field

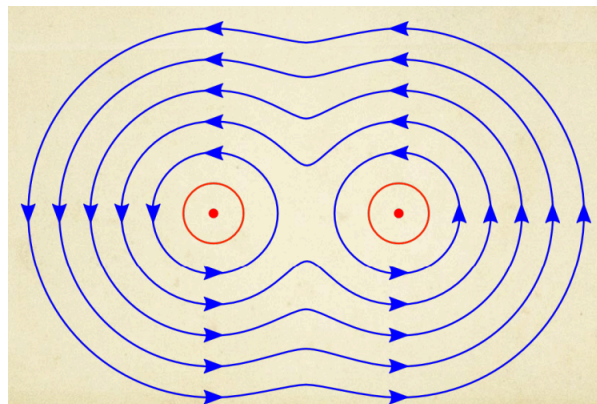
- A conductor that is carrying an electric current and lies at an angle to a magnetic field experiences a force.
- This is called the motor effect, and is defined as the force experienced by a current-carrying conductor in an external magnetic field.
 - The force is calculated by $F = lB\sin\theta$:



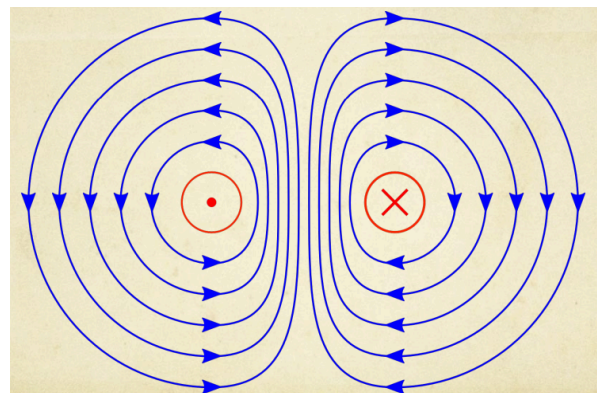
- The maximum force is produced when the current is perpendicular to the magnetic field.
 - The force decreases in magnitude as the current becomes more parallel to the magnetic field, reaching zero when they are fully parallel.

Parallel and Antiparallel Currents

- Currents produce magnetic fields, and two currents will induce magnetic fields that induce a force on the other current.
- The direction of the currents may be parallel or antiparallel.
 - Parallel means flowing in the same direction and antiparallel means flowing in the opposite direction.
- Parallel currents attract each other.



- Antiparallel currents repel each other.

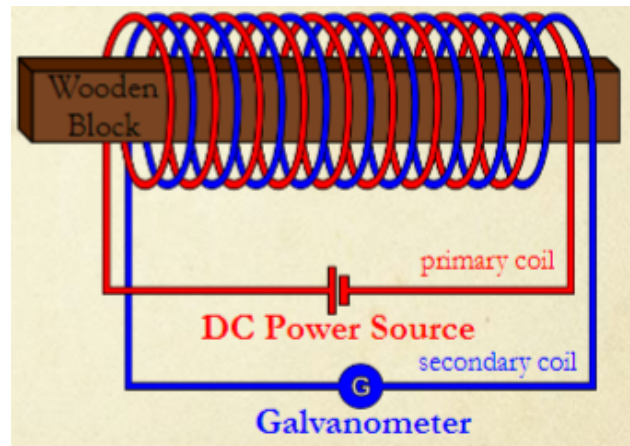


- The force per length between two currents is given as $\frac{F}{l} = \frac{\mu}{2\pi} \frac{I_1 I_2}{r}$, where force per length is in newtons per metre (Nm^{-1}).

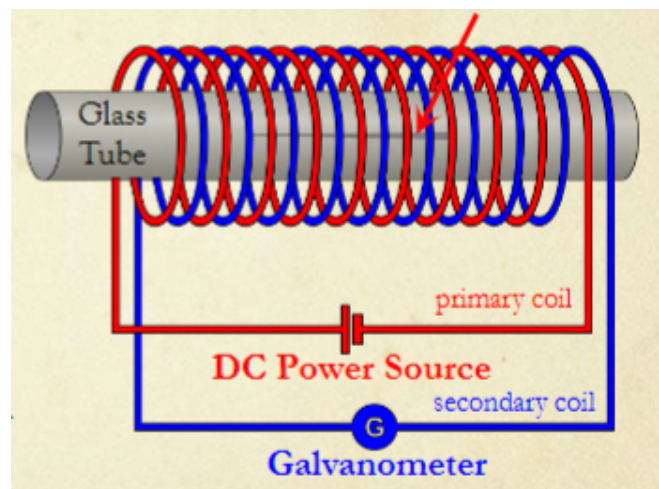
Electromagnetic Induction

Relationship Between Current and B Field

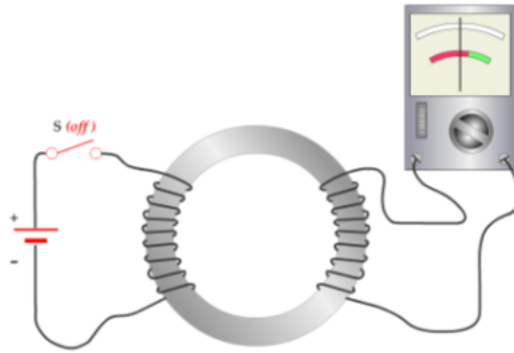
- Hans Christian Ørsted (1777-1851) discovered the first link between electricity and magnetism.
 - This occurred when he passed a current through a wire and noticed a nearby compass deflecting.
- The idea that the inverse, that is, a magnetic field could produce electricity was studied by many scientists.
 - This included Humphry Davy and William Hyde Wollaston.
 - Most notably, Michael Faraday performed several important experiments to produce a current from a magnetic field.
- Faraday wound copper wire around a piece of wood, and wound a second copper wire between the wires of the primary coil.
 - When the power source was turned on, the galvanometer measured a current pulse.
 - When the power was left on, no current was measured.
 - When the power source was turned off, the galvanometer measured a current pulse in the opposite direction to the previous pulse.



- The experiment was repeated, but the wood was replaced with a glass tube with a steel needle inside of it.
 - The same results were recorded, with the current pulses being slightly larger.
 - The needle also became magnetised.



- In order to reinforce the concept of electromagnetic induction, Faraday completed another experiment.
 - Faraday wrapped separate coils around opposite sides of an iron ring.
 - The results were identical to the previous experiments, but the currents measured were much greater.
 - This highlighted the phenomenon of mutual induction.

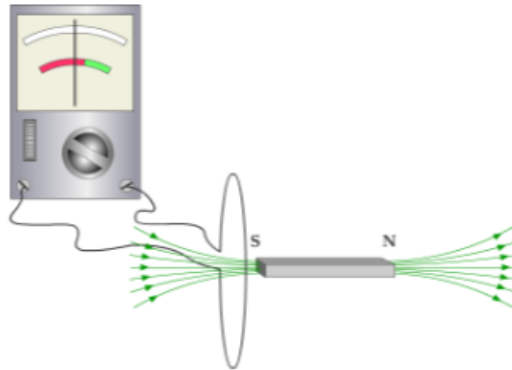


- The nature of Faraday's experiment was the changes in current caused by turning the power source on and off.
 - This resulted in a changing magnetic field associated with the primary coil.
 - This changing magnetic field induces a current in the secondary circuit.
- This meant that the fundamental factor in electromagnetic induction is a changing magnetic field.

Electromagnetic Induction and EMF

- While it was known that a changing magnetic field induces a current in a conductor, it was unknown what caused the current to flow.
- An electric potential that drives charge motion was suggested, and was referred to as electromotive force or EMF.
 - EMF is also denoted as ϵ and is measured in volts (V).
 - EMF is not an actual force, but an electric potential.
- Faraday completed another experiment by moving magnets through solenoids.
 - When he moved the magnet towards the coil, a current was measured in the solenoid.
 - When the magnet was held still, no current was measured.
 - When the magnet was moved away from the coil, the current in the solenoid was produced in the opposite direction to previously.

- He also found that the faster he moved the magnet, the greater the currents produced.



- These observations eventually led to Maxwell's development of his quantitative electromagnetic theory.

Flux and Flux Linkage

- A magnetic field is represented by field lines.
- Magnetic field strength can be considered as the number of field lines per unit area.
 - Magnetic field strength is denoted as B and is measured in teslas (T).
- Magnetic flux is the number of field lines passing through a specified area A .
 - Magnetic flux is denoted as Φ and is measured in webers (Wb).
- The relationship between magnetic flux and magnetic field strength is:

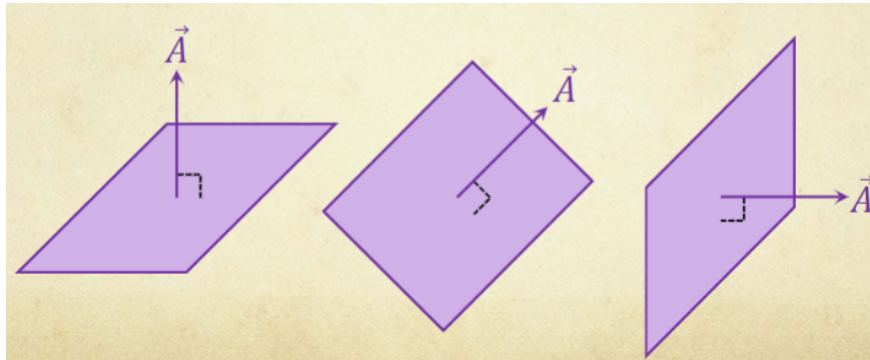
$$\Phi = BA \text{ and } B = \frac{\Phi}{A}$$

- This means that when magnetic field strength or the area specified increases, magnetic flux also increases.
- When multiple loops are connected in series in a solenoid, each loop experiences the same flux as a single loop would.
 - However, the flux of each loop is linked to the next, where the total flux through the solenoid is called flux linkage.
 - Flux linkage is denoted as λ and is measured in webers.

- The formula for flux linkage is below:

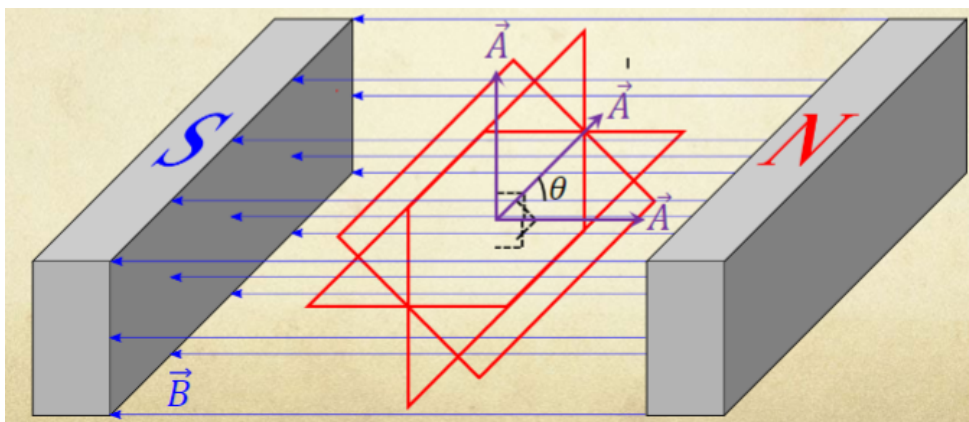
$$\lambda = N\phi = NBA$$

- The vector of A is represented as the normal to the area, as shown below:



- This is important when considering a rotating loop of area A in a magnetic field.
 - The magnetic flux is at a minimum when $\theta = 90^\circ$ and the vector of A is perpendicular to the vector of B.
 - The magnetic flux is at a maximum when $\theta = 0^\circ$ and the vector of A is parallel to the vector of B.
 - The formula for the magnetic flux with regards to θ is below:

$$\phi = BA \cos \theta$$



Faraday's Law and Lenz's Law

- Faraday's Law states that an induced EMF is negatively proportional to the rate of change of the magnetic flux.

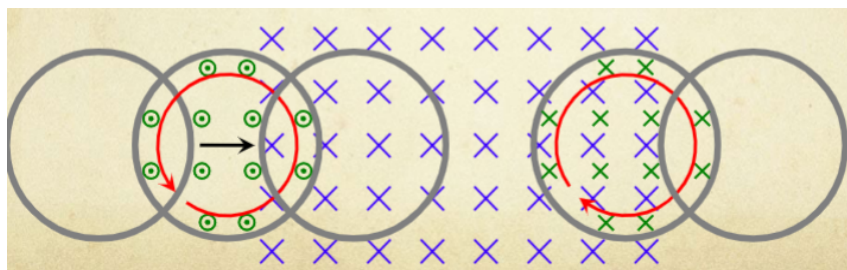
- This is represented below, where N is the number of turns in the solenoid.

$$\varepsilon = - N \frac{\Delta\phi}{\Delta t}$$

- Changes in magnetic flux can be due to changes in N, B, A or θ .
- Lenz's Law states that an induced EMF always gives rise to a current whose magnetic field opposes the original change in magnetic flux.
 - This means that the net result of the induction acts to oppose the cause of the change in magnetic flux.
 - This explains the negative sign in Faraday's Law, and arises from the Law of Conservation of Energy.
- The Law of Conservation of Energy is applied because if the result of the induction does not oppose the original change, it will keep inducing currents and magnetic fields until their values reach infinity.
 - This does not obey the Law of Conservation of Energy.

FECMO

- FECMO is an mnemonic that details the process of induction in order.
 - F stands for change in flux, E stands for EMF induced, C stands for current induced by the EMF, M stands for magnetic field induced by the current, and O stands for how the current and magnetic field oppose the initial change in flux.
- An example where FECMO can be applied is when a conducting loop is moved through a magnetic field.



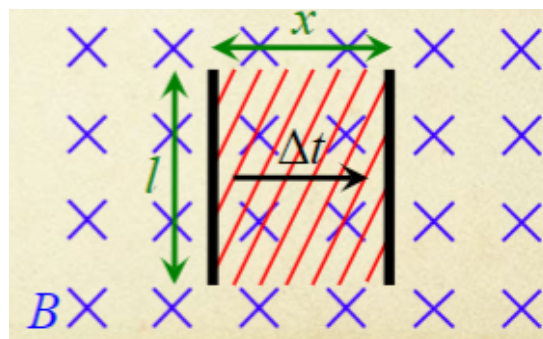
- The first instance of magnetic flux change is when the wire is entering the field.

- The change of flux induces an EMF according to Faraday's Law.
- This EMF induces a current opposing the increasing magnetic flux.
- This current induces a magnetic field that is in the opposite direction to the original magnetic field.
- The second instance of magnetic flux change is when the wire is entering the field.
 - The change in flux induces an EMF according to Faraday's Law.
 - This EMF induces a current opposing the decreasing magnetic flux.
 - This current induces a magnetic field that is in the same direction to the original magnetic field.

Isolated Wires in B Field

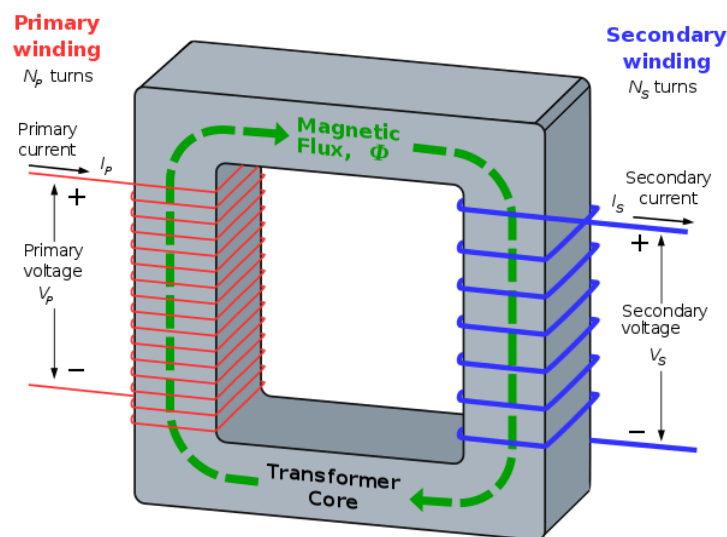
- When an isolated wire moves through a magnetic field, the motion of charges result in a force from the motor effect.
 - This motion of charges results in a separation of charge, producing a voltage and generating an electric field.
 - This electric field builds up, resisting the motion of the charges.
- A moving wire experiences a changing flux.
 - If the wire has the length l and moves a distance x , the formula for EMF is shown below:

$$\varepsilon = - Blv$$



Transformers

- Transformers are used to change AC voltage between electrical circuits.
- They are used in electricity grids and are essential to the transmission of electricity around the city, state and country.
- They are also used in many household appliances.
 - These include high voltage appliances such as cathode ray televisions, or low voltage appliances such as computers.
- The transformer has three main components.
 - The primary coil is the solenoid carrying input voltage.
 - The secondary coil is the solenoid carrying output voltage.
 - The iron core magnetically links the primary and secondary coils.



Function of Transformers

- Changes in current in the primary coil induces a changing magnetic field, which creates a changing magnetic flux in the iron core.
 - An alternating current is required for this to work because a direct current will not induce a change in flux.
- This magnetic flux induces an EMF in the secondary coil, inducing a current.

- In an ideal transformer all electrical energy is transferred from the primary coil to the secondary coil.

Voltage and Current Ratios

- The relationship between the voltages, currents and the number of turns of the primary and secondary coils is below:

$$\frac{V_p}{V_s} = \frac{N_p}{N_s} = \frac{I_s}{I_p}$$

Types of Transformers

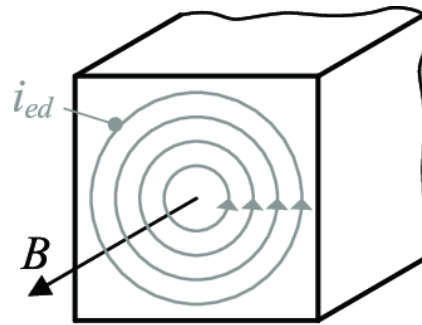
- Transformers can be classified according to the voltage ratio.
- A step-up transformer has a greater secondary voltage than primary voltage.
 - This means that the secondary coil has more turns than the primary coil.
- A step-down transformer has a greater primary voltage than secondary voltage.
 - This means that the primary coil has more turns than the secondary coil.
- The comparison in voltage, number of turns and current between step-up transformers and step-down transformers are summarised below:

	Step-Up	Step-Down
Voltage	$V_p < V_s$	$V_p > V_s$
Number of Turns	$N_p < N_s$	$N_p > N_s$
Current	$I_p > I_s$	$I_p < I_s$
Power	$P_p = P_s$	$P_p = P_s$

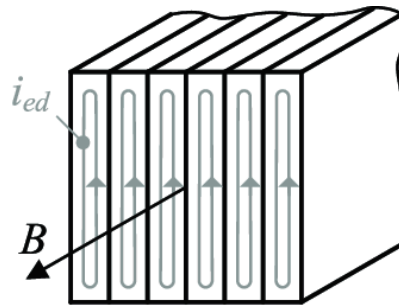
Energy Losses in Transformers

- In reality, transformers are not ideal and do not transfer all energy between the coils.
- The AC current in the primary coil produces an alternating magnetic field, with the changing magnetic flux produced leading to eddy currents.

- These eddy currents allow energy loss in the form of resistive heating.



- In order to minimise energy losses in the core, eddy currents need to be minimised whilst allowing magnetic flux to propagate.
- This can be done through laminated cores, which is when layers of iron are alternated with thin non-conducting layers.
 - This forms more eddy currents, but they are much smaller and cause significantly less energy loss.



Power Loss and the Transmission of Electricity

- Electricity is usually generated far from cities, so they need to be transmitted over large distances across a large national grid of transmission and distribution lines.
- These wires have low resistivity but resistance builds up over hundreds of kilometres.
 - This electrical resistance results in energy loss in the form of heat, and is referred to as line loss.
 - Line loss is represented in the formula below:

$$P_{\text{loss}} = V_{\text{lines}} I \text{ or } P_{\text{loss}} = I^2 R_{\text{lines}}$$

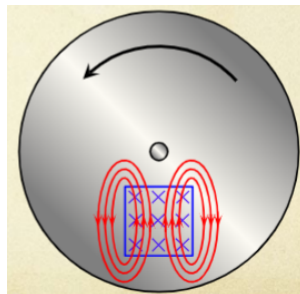
- The formula above shows how power loss is best minimised by reducing current.
- Transformers can be used to reduce these losses by increasing the voltage, which decreases current through the voltage-current ratios.
 - This allows electricity to be transmitted over large distances with small energy losses.
- Transmission lines have high voltage and low current, and distribution lines have low voltage and high current.
 - This means that step-up transformers are used for transmission and step-down transformers are used for distribution.

Transformers in Household Appliances

- Some electrical appliances in the home use a transformer.
 - Appliances such as a cathode ray television and microwaves require high voltages and therefore need a step-up transformer.
 - Appliances such as an alarm clock or a mobile phone charger require low voltages and therefore need a step-down transformer.

Eddy Currents in Solid Conductor

- When a section of continuous solid conductor passes through a magnetic field, a current is induced.
 - This causes charge separation between the inner and outer regions of the area affected by the magnetic field.
- This charge separation causes current loops to develop, which are also called eddy currents.

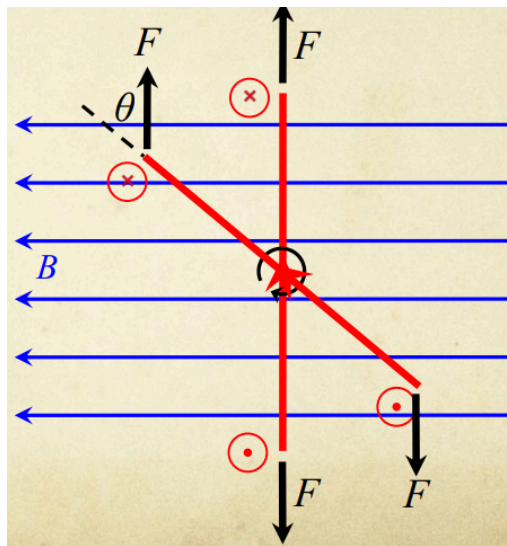


- Eddy currents convert kinetic energy to electrical energy by using the movement of the conductor to induce a current.
 - This means that eddy currents are used in electromagnetic braking, as eddy currents convert kinetic energy to electrical energy, slowing down the rotation of the wheel.
- Electrical energy can also produce resistive heating, which is used in induction cooktops.
 - Eddy currents induced in pans can heat pans directly by resistive heating.

Applications of the Motor Effect

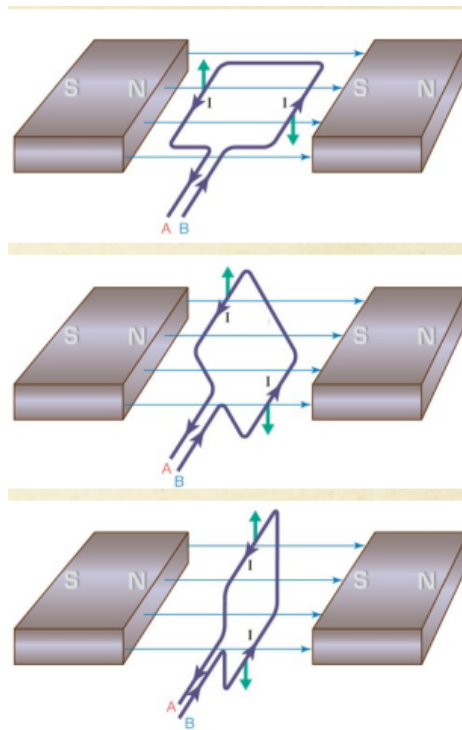
Torque in a Magnetic Field

- Torque on a current-carrying coil as it rotates in a magnetic field can be represented by the formula $\tau = IAB\sin\theta$.
 - If there are n loops of wire in the coil then it is represented by $\tau = nIAB\sin\theta$.

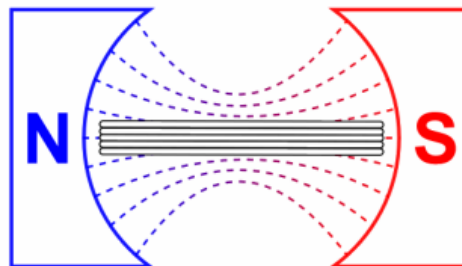


- Maximum torque occurs when the area vector of the loop is perpendicular to the magnetic field, and minimum torque occurs when the area vector of the loop is parallel to the magnetic field.

- An intermediate torque occurs in between 0° and 90° .



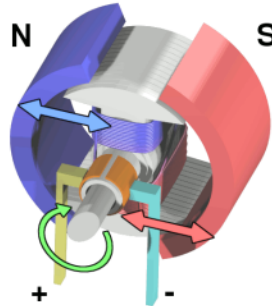
- Magnets are used to provide the magnetic field for the motor effect to happen.
 - They can be either permanent magnets or electromagnets.
- The magnets can be arranged to produce a radial field.
 - This ensures that the area vector and the magnetic field are perpendicular for a larger range.
 - It also allows for more consistent rotation of the coil.



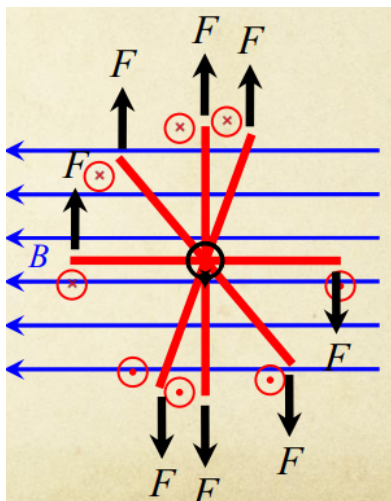
Motors and Generators

- Electric motors transform electrical energy to kinetic energy.

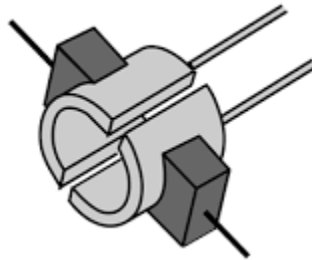
- They can be classified according to their power source.
 - Motors can either have an alternating current (AC) power source or a direct current (DC) power source.



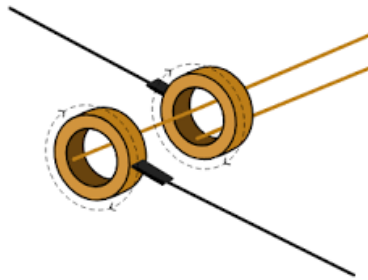
- The main features of DC motors are the armature, commutator and brushes.
- The armature is the coils of wire placed in the magnetic field, and are the current-carrying loop for the motor effect.
 - Multiple loops are used and they are usually wound around an iron core to increase the motor effect.
- Brushes allow continuous electrical contact between the rotating armature and the stationary circuit.
 - They must have sliding contact with the armature.
- The commutator is required to allow the armature to turn completely.
 - As the coil turns in the magnetic field, the torque decreases to zero.
 - The coils continue to turn due to inertia, but torque will return the coils to the perpendicular position.



- DC motors require a split ring commutator, which allows constant current to be converted into circular motion.
 - The ring on the shaft with two gaps allows the reversal of polarity, and maintains the direction of torque.



- AC motors do not require a reversal of polarity, so a slip ring commutator is used instead.

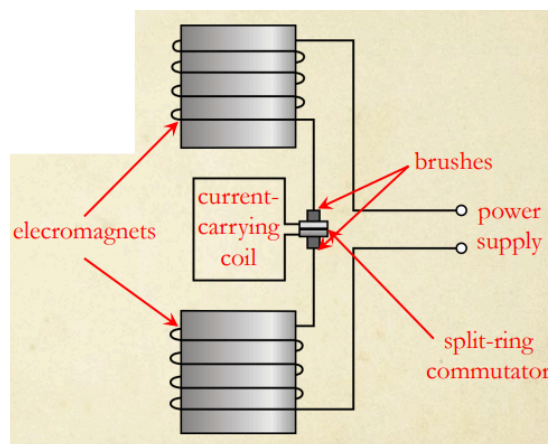


- Electric generators transform kinetic energy to electrical energy.
 - Similarly to electric motors, they can be classified according to the power they generate.
- The main features of generators are identical to their corresponding motor.
 - The magnets are known as the stator and the armature and commutator are known as the rotor.
- A table comparing the mechanisms and parts of motors and generators is below:

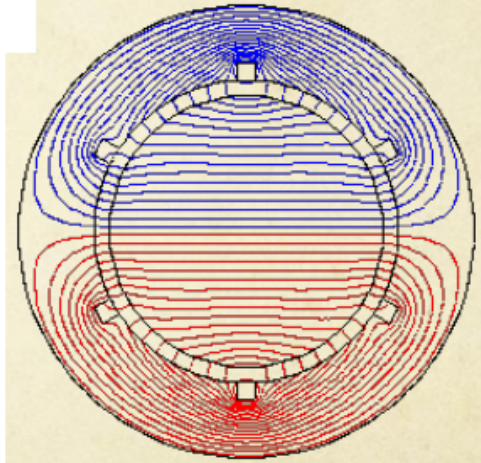
	Motors	Generators
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Magnets	✓	✓
Armature	✓	✓
Commutator	✓	✓
Brushes	✓	✓
Input Energy	Electrical	Kinetic
Output Energy	Kinetic	Electrical
Mechanism	Motor Effect	EM Induction

- A universal motor works with AC and DC power sources.
 - The design is based on a DC motor with electromagnetic provided for the magnetic field.

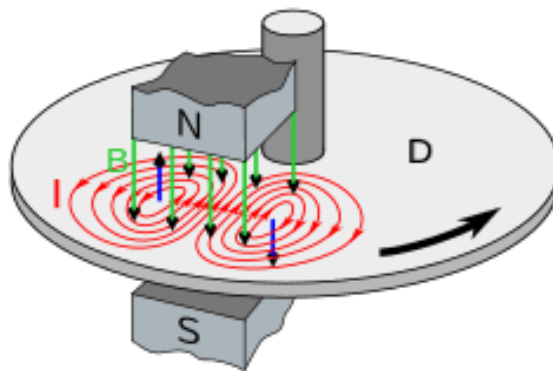


- In a 3-phase induction motor, three independent AC voltages are out of phase by 120° .
 - The coiling is designed to form three pairs of opposing poles.
 - As alternating currents phase in and out, the poles rotate around outside.
 - The changing flux induces eddy currents which oppose the change in flux.
 - Lenz's Law acts to minimise the relative motion in the motor.



Other Uses

- In electromagnetic braking, the relative motion between the conductor and the magnetic field results in induction.
 - The change in flux induces an EMF, further inducing currents in the form of eddy currents.
- The Law of Conservation of Energy comes into play as electromagnetic induction increases the electric energy, decreasing kinetic energy.
 - Electrical energy is transformed into kinetic energy.



- In induction cooktops, a high frequency AC electric coil produces an oscillating magnetic field.
 - This induces eddy currents in the base of the pan, which heat up due to resistive heating.

- Resistive heating can be calculated by $P = I^2 R$.

